

ML Specification Report — playground-series-s3e20

Rwanda CO2 Emission Prediction · our from-scratch agent (structural location-week empirical-Bayes model + tree-search)

Figures grounded in the competition's `config.yaml`, `facts.json`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`, with external benchmarking from `/tmp/ext_facts.json`.

Overview

The task is to predict weekly CO2 emission over Rwanda from satellite-sensor readings (metric: RMSE, lower is better). Unlike our other episodes, this is the one competition where **structure completely dominates**: a purely structural per-(location, week) empirical-Bayes historical-mean model beats every gradient-boosted tree we throw at it. Our committed champion is that structural model — no GBDT, no deep net — scoring **CV RMSE $\approx$ 21.15** (Leave-One-Year-Out); a tree search then nudges the composed structural optimum to **21.0589**. The NVIDIA reproduce-agent, applying the technique from a strong public kernel (kacperrabczewski/rwanda-co2-step-by-step-guide), reaches **19.648** — NVIDIA ahead.

Why it matters. Satellite-based emission estimates support climate-policy monitoring and carbon accounting where ground sensors are sparse — a pressing need across developing regions such as Rwanda.

Data

Purpose of Data. Predict the continuous `emission` target for each (latitude, longitude, year, week) cell in Rwanda, from satellite-derived atmospheric measurements — a **regression** Playground Series episode (S3E20). **Data Format** is clean **tabular CSV**, entirely numeric. **Data Volume** is **79,023 training rows / 24,353 test rows**, with **74 numeric features** and **0 categorical features** once the `ID_LAT_LON_YEAR_WEEK` key is set aside. The single most important structural fact from EDA is that the training set is a **perfect balanced panel** — **497 locations $\times$ 53 weeks $\times$ 3 years = 79,023 rows** — so every (location, week) cell is observed in every training year (2019–2021), and the test set is the same grid for 2022.

Data Quality is mixed. There are **0 duplicate rows**, but missingness is heavy and structured: the entire `UvAerosolLayerHeight_*` **sensor group is 99.4% missing** (78,584 / 79,023 rows), `NitrogenDioxide_*` is $\sim$ 23% missing (18,320 rows), and `SulphurDioxide_*` $\sim$ 18% (14,609 rows). Several sensor-geometry columns also show a large **train $\leftrightarrow$ test mean shift** (e.g. `SulphurDioxide_sensor_azimuth_angle` -21% , `Formaldehyde_sensor_azimuth_angle` -135%), a warning that raw sensor angles do not transfer across the year boundary. The target is **extremely right-skewed** (`emission` mean 81.94, median 45.59, min 0.0, max 3167.77, **skew 10.17**, **kurtosis 157.6**), which motivates a `log1p` transform in the tree members. Crucially, the **satellite features carry almost no linear signal**: the strongest target correlation is merely `longitude` (Pearson 0.103 / Spearman 0.291), and every sensor column has $|\text{Pearson}| < 0.07$. The predictive signal lives

in *which cell and which week* a row belongs to, not in the sensor readings — the finding that shapes the whole solution.

Annotation Guidelines. The label is the continuous `emission` value (≥ 0); submissions are real-valued predictions scored by RMSE against the held-out 2022 grid. The provided `sample_submission.csv` baseline is simply the **mean of the training target**.

Feature Set. The champion is **structural, so its “features” are the panel key, not the sensors**: it predicts each test cell by an empirical-Bayes-shrunk historical mean of `emission` over that cell’s (location, week) group across the training years, with COVID-year down-weighting, neighbor-week smoothing, and per-year weighting (see the Models section). The 74 raw sensor columns enter only the **GBDT diagnostic** (63 sensor columns + latitude/longitude/week, log1p target); the earlier tree tiers additionally used loc-week and loc target encodings on top of the sensor block. Sensor-derived features were found to add essentially nothing over the structural key.

Splitting strategy. The canonical scheme is **Leave-One-Year-Out (LOYO) CV, 3 folds (hold out 2019 / 2020 / 2021 in turn), seed = 42**. This is the honest analogue of the train → test year gap (train 2019–2021 → test 2022) and replaced the early time-based split (2019–2020 train / 2021 val) used in v1. Note that the automated `validation_hint` suggested a plain KFold (“continuous i.i.d. target → standard KFold”); we overrode it because the task is a per-year panel, and LOYO is the only split that measures cross-year generalization. **The leaderboard is closed and this run is OOF-only** — no test predictions were generated for the structural champion, so all scores here are **LOYO CV**; the `leaderboard` field is `null` (**not recorded**) and no public/private LB score exists for the committed champion.

Models & Architecture

Purpose of Architecture. A regressor minimizing RMSE on 2022 Rwanda CO2 emission. **Architecture Type** is — honestly — **a structural, non-parametric empirical-Bayes historical-mean model over the (location, week) panel**, *not* a deep network and *not* a GBDT. The mapping to a DL/GBDT architecture is therefore **N/A**; the one GBDT we fit is retained only as a diagnostic (below).

Input Format for the champion is the panel key (location, week, year) rather than a dense feature vector, so a conventional **Input Dimension** count is **N/A** — the model consumes group identity, not the 74 sensor columns.

Architecture Description. For each (location, week) cell the model forms a historical mean of `emission` across the training years and shrinks it toward broader means via empirical-Bayes (the committed root uses shrinkage α , COVID-year down-weight `w2020`, neighbor-week smoothing weight `wnb`, and a smoothing window). The tree-search champion adds a per-year weighting axis. The committed Phase-B root config (tree node #0) and the tree-search optimum (node #28) are:

Axis	Root (node #0)	Tree-search optimum (node #28)
EB shrinkage <code>alpha</code>	1.0	0.99
COVID down-weight <code>w2020</code>	0.24	0.2
neighbor-week smoothing <code>wnb</code>	0.28	0.3

Axis	Root (node #0)	Tree-search optimum (node #28)
smoothing window	±1	±1
year_weights (2019/2020/2021)	— (none)	0.6 / 0.2 / 1.2
LOYO OOF RMSE	21.1487	21.0589

Model Complexity is not a trainable-parameter count (N/A for a structural model); it is **4 tuned scalar hyperparameters** in the root, plus the new 3-value `year_weights` axis at the optimum. **The GBDT diagnostic** — a single lightweight LightGBM (`n_estimators` = 400, `learning_rate` = 0.05, log1p target, 63 sensor + lat/lon/week features, LOYO CV; tree node #9) — scored **solo RMSE 51.40**, far worse than the structural model, re-confirming across five prior Phase-B runs that GBDTs earn ~0 blend weight against this signal.

Training procedures

There is no gradient-based training loop for the champion; “training” is (a) computing the shrunk historical means and (b) searching a handful of scalar weights. The improvement trajectory across our runs was:

Stage	Configuration	CV scheme	RMSE
v1	LGB+XGB+Ridge ensemble on sensor features	time-based (2019–20 / 2021)	38.51
v1-final	LGB+XGB 50/50 on sensor features	time-based	33.21
v2	LGB+XGB+CatBoost + loc-week target encoding	LOYO (3-fold)	28.34
v2+TE	+ pure structural TE member added to blend	LOYO	22.6488
Phase-B	structural TE tuned (EB shrinkage / COVID / neighbor smoothing)	LOYO	21.6317 → 21.1487 (root)
Tree-search	JOINT lineage: all 4 axes + new <code>year_weights</code> , co-optimized	LOYO	21.0589 (node #28)

Across v2 onward the blend weight on every GBDT member collapsed to 0 — the structural target-encoding member alone carried the score. The **Loss Function / Optimization Algorithm / Learning Rate / Learning Rate Scheduler / Batch Size** are all **N/A for the structural champion** (no gradient objective, no shrinkage schedule, no mini-batches); the only place these apply is the GBDT diagnostic, whose loss is L2 on the log1p target with `learning_rate` = 0.05 — and it loses decisively.

The tree search co-optimized the structural axes rather than Phase-B’s one-axis-at-a-time manual tuning: re-scanning the individually-tuned axes (`w2020` , `wnb`) in isolation only **confirmed** Phase-B’s optimum (an honest tie), but **JOINT-moving all axes together plus the new `year_weights` axis** reached a composed optimum (21.0589) that the manual protocol was structurally unable to find. The largest single-axis gain was the new `year_weights` idea (down-weight the anomalous 2019/2020 years, up-weight 2021 as closest to the 2022 test year), which alone moved 21.0951 → 21.0808.

Training Duration for the entire tree sweep was **~33 s of wall-clock** over **38 evaluated nodes** (`our_wall_s` = 33, `our_train_fits` = 38); an individual structural node fits in ~ 0.07 s (pure groupby, no model training). **Training Memory** was **not recorded** — a $79k \times 74$ float panel is trivial on the single arm64 CPU machine and peak was not measured. **Transfer Learning** is **N/A (no pretrained weights)**; its analogue is our cross-competition experience library that carried the “structure-first, GBDT-as-diagnostic” prior into this episode. **Data Augmentation** is **N/A (tabular)**; the structural analogues are neighbor-week smoothing and empirical-Bayes shrinkage.

Reproducibility Standards are strict: fixed LOYO fold seed = 42, digit-for-digit root verification (node #0 re-computed to 21.1487 before the search began), and a faithful-replay contract. The recorded replay is **38/38 nodes exact**, $\Delta = 0$ (`faithful_replay`), and the search state is marked `stopped` with `stop_reason` “faithful replay of committed tree”.

Inference procedures

Decision Threshold is **N/A** — RMSE is a regression metric, so predictions are the real-valued shrunk historical means (clipped to ≥ 0 in the earlier tree tiers), never thresholded. **Inference Duration** and **Inference Memory** were **not recorded**; structural scoring is a groupby lookup, effectively free on CPU. Note the champion search was **OOF-only** — no 2022 test predictions or submission file were produced (the competition is closed), so there is no submitted-inference profile to report.

Evaluation & Benchmarking

Task for Performance Evaluation. Predict 2022 weekly CO2 emission across the Rwanda grid; score by RMSE. **Performance Metrics.** Our committed structural champion reaches **CV RMSE ≈ 21.15** (LOYO), improving $33.21 \rightarrow 28.34 \rightarrow 22.65 \rightarrow 21.63 \rightarrow 21.15$ across the trajectory; the tree search then refines the structural optimum to **21.0589** (node #28). A sibling **BLEND node (#37)** mixed the structural node with the GBDT diagnostic and scored 21.0332 — but the GBDT earns only **2.17% weight** on 3-fold LOYO with the weight fit on the same OOF it is scored on, which `STATUS.md` explicitly flags as low-confidence, within CV-noise, **not a robust conclusion**. We therefore report the purely-structural **21.0589 / 21.15** as the honest result. No public/private LB figures exist for this champion (comp closed, OOF-only) — **not recorded**.

Performance Benchmarking. The comparator is the **NVIDIA reproduce-agent**, which applies the *technique* from a strong public kernel (`repro_type = technique`):

Agent	Approach	CV / kernel RMSE	Note
NVIDIA	reproduces technique of <code>kacperrabczewski/rwanda-co2-step-by-step-guide</code>	19.648	winner
Our agent	from-scratch structural location-week EB model + tree search	21.15	behind by ~7%

NVIDIA leads (lower RMSE is better): 19.648 vs our 21.15, a gap of ~ 1.5 RMSE ($\sim 7\%$ relative). The honest read: this episode rewards a specific, well-known structural recipe for the Rwanda panel that the public kernel encodes tightly; our from-scratch agent independently discovered the *same* structural regime

(historical means dominate, GBDTs get ~ 0 weight) and tuned it to 21.06–21.15, but did not close the last increment to the kernel’s optimized 19.648. The load-bearing finding — that a 4-scalar structural model *beats every GBDT ensemble* here, confirmed five separate times — is itself the report’s central result.

Final-Architecture Re-Run (v5 pipeline, Stage 0.5 in-lane) — 2026-08

- s3e20 is outside the frozen 20-competition baseline (late submission closed, no private-LB anchor), so it has no re-run lane.